Measuring the Hyperfine Structure of Rubidium Isotopes using Optical Pumping

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Abstract

The quantum structure of mulitelectron atoms is a complex field of study. Nuclear and atomic interactions are tiny and yet have large ramifications in our understanding of the basic units of matter and energy. The existence of quantized states in miniscule particles brings a new framework to atomic interactions, one that is observable using specialized techniques. This report details a method to observe and quantify the magnetic interactions between nuclei and their electron clouds, called the hyperfine structure. The method is called optical pumping, and it is used to measure the energy splitting g-factors of quantum states in rubidium isoptopes, with experimental results within a 3% percent difference of known values.

1 Introduction

The quantization of energy in fundamental particles has defined a new type of mechanics used in the 'quantum domain'. The description of nuclei and electrons falls in this realm and the distribution of energy states must be understood to model the physics of such systems. The basic principle is that a fundamental particle (or a nucleus, which we can consider as a particle) can only have certain energies, called states. These states are associated with different proprties of the particle, such as angular momentum or the magnetic moment. The most basic distribution of states comes from the energy itself, what are called the n states. n is called a quantum number and is associated with the energy at each state. These states are sufficent to understand the particle on its own, but the existence of external interactions affects energy, and therefore complicates how the states are distributed.

1.1 Quantum State Splitting

This report focuses on the outer electron and the nucleus of Rubidium. To a good approximation, multi-electron atoms can be described in terms of the nucleus and the electrons in the outer shell, since the inner electrons are 'full' and can be described as 'canceling each other out' [1]. Rubidium has one valence electron in the 5S shell, which can be described in terms of angular momenta [2]:

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \tag{1}$$

$\mathbf{L}$ is the angular momentum of the electron around the nucleus, $\mathbf{S}$ is the intrinsic angular momentum, called the spin, of the electron, and they sum to form $\mathbf{J}$ the total angular momentum.

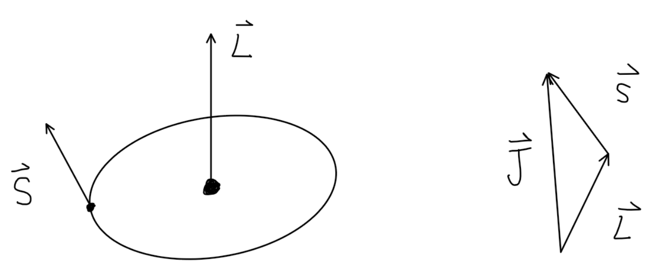

Figure 1: angular momenta of the outer electron of a rubidium atom

Now, this electron is said to be in the s sublevel but it can be excited to the p sublevel as well. The S level is the electronic ground state of the atom, in this state $\mathbf{L} = 0$, and the total angular momentum can only take on one value, the value of $\mathbf{S}$. The first excited state, the P state, $\mathbf{L} = 1$, and the total angular momentum can take on the values $L + S$ and $L - S$, two states. Thus the original P state, defined by the energy, splits into two states, $P_{1/2}$ and $P_{3/2}$, that are created due to the energy contributed by the orientation of the angular momentum. This splitting defines the **Fine Structure** [1] of the energy state distribution.

This angular momentum contributes further energy when coupled with the angular momentum of the nucleus. Similar to the electron, the nucleus has an intrinsic angular momentum (spin) and therefore has a magnetic moment. Two magnetic moments will interact with each other, and that interaction has energy based on the relative orientations of the moments, so in other words the spin of the nucleus will couple magnetically with the spin of the electron creating a total angular momentum of the atom, called $\mathbf{F}$ [1].

$$\mathbf{F} = \mathbf{I} + \mathbf{J} \tag{2}$$

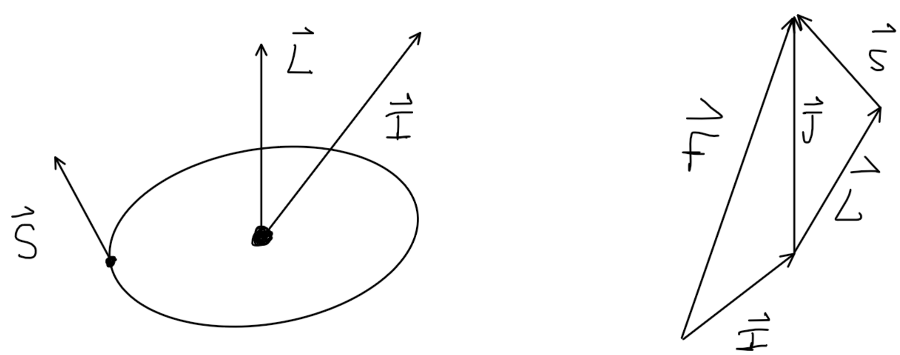

Figure 2: magnetic coupling of the nucleus and outer electron in a rubidium atom, creating a total angular momentum of the atom itself, $\mathbf{F}$

The energy contributed from the magnetic interaction is based on the orientation, which can be in multiple different states, thus splitting the energy states further. This new distribution is called the **Hyperfine Structure**.

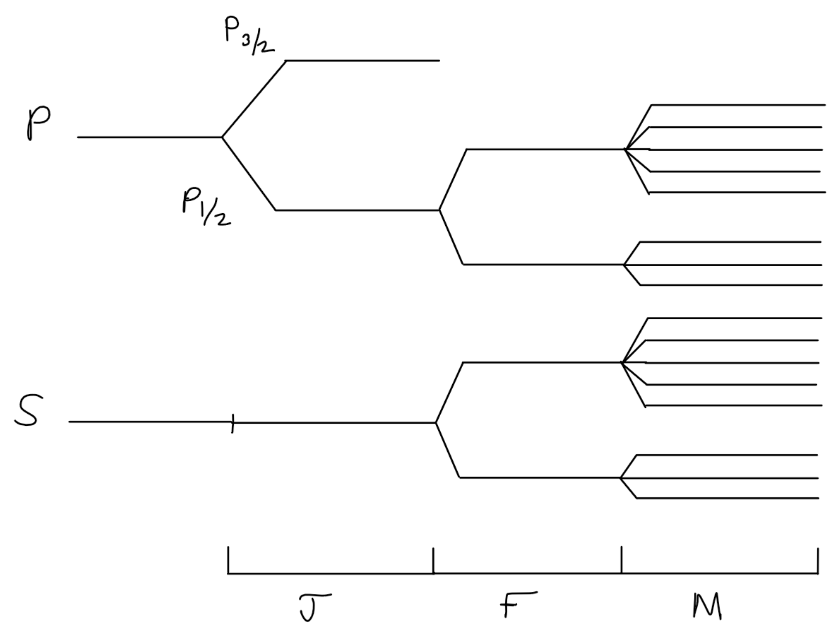

Figure 3: state splitting of the rubidium atom due to angular momentum orientation and magnetic moment dipole interactions, showing the J (fine) states, F (hyperfine) states, and M (Zeeman) states

Finally, if an external magnetic field is applied, the energies are split even further as the orientation of the magnetic moment determines contributed energy due to interaction with the external field. This is called the **Zeeman Splitting**. The hyperfine states split into $2F + 1$ states quantized by a number m.

The interaction energy associated with the hyperfine splitting is given in the following equation [1]:

$$W = g_F \mu_0 B M \tag{3}$$

To find the difference in energy (meaning the split), it is simply

$$\Delta E = W(M + 1) - W(M) = g_F \mu_0 B \tag{4}$$

g_F is the **Lande g-factor** associated with the hyperfine splitting, and will be experimentally determined in this experiment. The rubidium sample provided contains two isotopes, ^{87}Rb and ^{85}Rb. The calculation of the g-factors requires the knowledge of the quantum numbers:

$$g_J = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)} \tag{5}$$

$$g_F = g_J \frac{f(f+1) + j(j+1) - i(i+1)}{2f(f+1)} \tag{6}$$

where g_J is the Lande splitting factor for the Zeeman splitting effect. The ground state for these isotopes have $l = 0$ and $s = \frac{1}{2}$, meaning $j = \frac{1}{2}$. ^{87}Rb has an intrinsic spin of $i = \frac{3}{2}$ [1], meaning $f = 2$, while ^{85} has a spin of $i = \frac{5}{2}$ [1] meaning $f = 3$. These result in the following g-factors:

$$^{87}\text{Rb}: \qquad g_F = \frac{1}{2} \tag{7}$$

$$^{85}\text{Rb}: \qquad g_F = \frac{1}{3} \tag{8}$$

These are the theoretical factors that we will attempt to observe in this experiment.

1.2 Optical Pumping

To observe the hyperfine structure in an atom, one must observe energy transitions between states in order to find the energy itself. This is accomplished through **resonance**. To excite a particle to a higher energy state, a photon with the exact energy difference to that state must strike the particle and be absorbed.

The classic model for describing optical pumping consists of three energy states, A, B, and C [3]. One can imagine three states, in which A and B are close together, with C much higher than the two. This process will be shown in figure 4. Now, the system begins with particles distributed among the states in accordance with thermodynamics, and we suppose that no particles are in state C at equilibrium. Two different energies are required to excite particles from the A and B states to C, so suppose a light beam hits the system, exciting particles. Suppose the photons containing the energy required to excite from B to C are filtered out, so that particles in state B cannot be excited. A particle in state A will be excited to C, and then either drop to A or B due to thermodynamic relaxation. Any particle that drops to A will be almost immediately excited back to C to repeat the process. A particle that drops to B is then unable to be excited to C and therefore remains in that state.

After some amount of time, all the particles must be in state B, and this is detectable since the photons will then no longer be absorbed into the material and instead pass straight through. The sample will be transparent with respect to that energy of light.

The challenge remains, then, to detect the 'pumped state', as after the initial pumping, the sample remains at an optically pumped state forever. This can be seen as a constant intensity curve on the oscilloscope, the sample is always transparent. However, by applying a radio-frequency magentic field, the resonant photons that match the energy between A and B can be used to deexcite the particles in the B state back to the A state, where they are again absorbed. This lowers the intensity of the light making it through the sample, which is detectable on the scope. This sharp dip represents a constant deexcitation from B to A, and then reexcitation from A to C, in a cycle. The magnetic field at which these radio frequency photons are created then gives insight into the desired energy splitting.

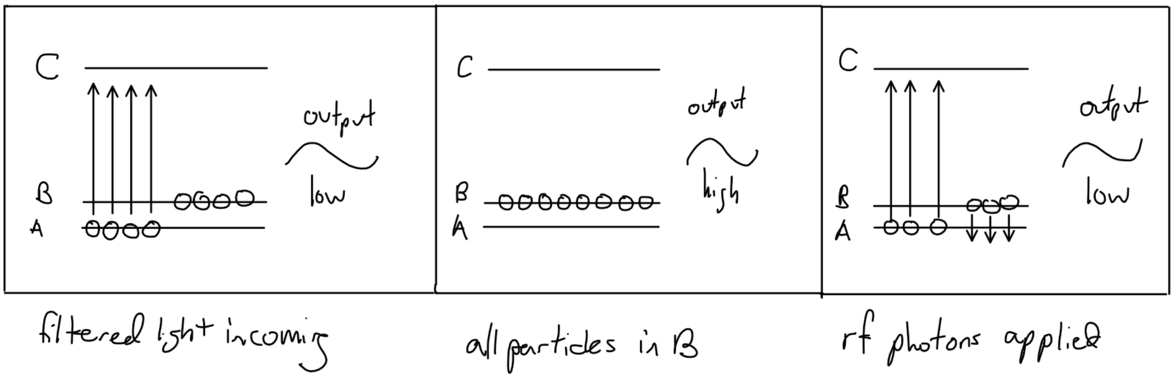

Figure 4: optical pumping process, including observation via radio frequency photons

In this experiment, the atoms in a lower J state will be excited to a higher J state, and then drop into the highest M state in the lower J state. The energy to excite atoms from this state is filtered out, so atoms will remain in this state until the resnonat frequency magnetic field is applied and they slip into lower states, and are excited again. This is where equation 4 comes into play, as we are find the magnetic field required to drop one M state.

The resonant frequency is related to the energy of the incoming frequency by Planck's equations:

$$\Delta E = hf_0 \tag{9}$$

and thus the governing equation that will be used to find the g-factors can be derived from equations 4 and 9:

$$g_F = \frac{h}{\mu_0}\frac{f_0}{B} \tag{10}$$

This equation demonstrates that a linear relationship between f_0 and B can be used to find the g-factor.

2 Experimental Procedures

2.1 The Optics

An in depth discussion of the optical setup is necessary to understand the processes taking place during the experiment.

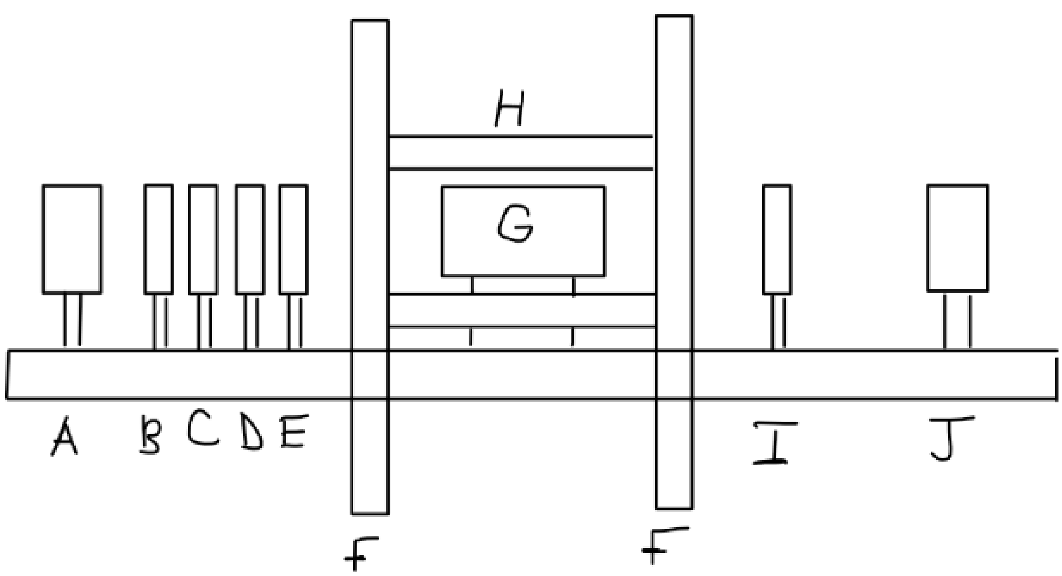

Figure 5: Optical rail including Helmholz coils (F & H), the lamp (A), and various filters

The optical rail is shown in figure 5, and each part is assigned a letter. The process begins with the discharge lamp, labelled 'A' in figure 5, where a light beam is emitted towards the sample. It passes through a convex lens (B), which is included to focus the light onto the rubidium sample itself. The next filters accomplish the task of creating a monochromatic, circularly polarized beam of light that will excite rubidium atoms. The filter labelled 'C' is an interference filter that only allows a 795 nm line to be transmitted to the sample, allowing excitations from only specific states. After the interference filter, the light passes through a linear polarizer (D) and a quarter plate (E) that circularly polarize the light, in order to prevent certain M state transitions. The light then passes into the temperature controlled oven (G). The temperature must be controlled to maintain thermodynamic equilibrium, and the oven contains the rubidium sample to be pumped. After the light passes through the oven and the sample, some is absorbed during the pumping process and due to the resonant field. The light that makes it through is sent through another convex lens (I) to widen the beam for detection by a photodiode (J).

The photodiode detector sends light intensity information back to the control panel for amplification and observation. In figure 5, objects F and H are Helmholtz coils for generating the horizontal and vertical magnetic fields respectively. Coil F also has an extra coil wrapped around it for the sweep field, which will be discussed in the next section.

2.2 Magnetic Fields

This experiment includes the generation of four different magnetic fields to affect the sample and create the precision necessary to optically pump the rubidium. To alter the resonance frequency (although this may not be necessary) a horizontal field can be turned on (parallel to the optical axis in figure 5) which acts as a static external field for the sample. The resonance frequency is then proportional to this field. During the experiment in this report, this field was never turned on.

A static vertical field could also be produced, by Helmholtz coils aligned with the sample, which could be used to cancel the Earth's magnetic field. The Earth's field could alter the excitation energy and therefore lower the accuracy of the experiment. The vertical field could be used to remove the vertical component of this field.

The radio frequency photons were generated in the form of an oscillating vertical magnetic field (perpendicular to the optical rail in figure 5). This field was used to deexcite the atoms from the pumped state, creating a measurable dip in light intensity.

The last field is the horizontal sweep field. This magnetic field swept through a predetermined range of field strengths so as not to overload the resonance state of rubidium. It steps through different strengths, creating a noticeable effect when it sweeps past the magnetic field at which the radio frequency is the resonance frequency. In this experiment, the radio frequency was kept constant and the magnetic field that matched with it was measured using the sweep field.

2.3 Measurment and Procedure

The first measurable effect was called **Zero Field Resonance**. In essence, when the magnetic field inside the sample was exactly zero, a resonance dip could be observed due to the nonexistence of the zeeman splitting effect. The observation of this signal allowed us to measure the horizontal component of the Earth's magnetic field at that point, and also to adjust our setup so that the Earth's field could not interfere with the experiment. This was accomplished by making sure the zero field resonance dip was as thin as possible, by both adjusting the vertical magnetic field and rotating the optical rail.

After the positionings were optimized, the radio frequency magnetic field was turned on and resonance dips were observed on the oscilloscope. Each peak corresponded with a specific isotope of rubidium, and the field needed to be measured. The magnetic sweep field operated by supplying current to the Helmholz coils, and this current flowed through a 1 ohm resistor. The voltage across this 1 ohm resistor was measured and, due to that resistance, was equal to the current. The magnetic field could then be calculated using the Helmholtz equation as [1]

$$B = 8.991 \times 10^{-3} IN/R \tag{11}$$

in units of gauss. I represents the current, N represents the number of turns on each side, and R represents the mean radius of the coils. The Earth's field, as calculated from the zero field resonance, was then subtracted from this value for a more accurate result.

The apparatus could also be used to measure the time it took to optically pump the sample. In order to do this the radio frequency field needed to be periodically turned off to allow the pumped sample to relax back to thermodynamic equilibrium. The rf field was modulated by a 5 Hz square wave that periodically turned on and off the field to allow us to observe the actual process of optical pumping.

3 Results and Analysis

3.1 The Earth's Field

As stated in the previous section, the magnitude of the Earth's magnetic field was obtained by measuring the location of zero field resonance. The vertical component was determined to be 0.335 ± 0.001 Gauss, and the horizontal component was measured as 0.351 ± 0.001 Gauss, giving a total magnitude of 0.35 ± 0.01 Gauss. This was compared with an expected value [4], and found to be similar but with a 25% difference (the expected value was 0.46 Gauss). The horizontal field was also used to offset the measured values of the resonances for rubidium to obtain a more accurate value.

3.2 Optical Pumping of Rubidium

The goals of this experiment were to observe optical pumping and to determine experimentally the Lande g-factors for the hyperfine structure of rubidium. This was done by applying a radio frequency and observing which magnetic fields forced the resonant frequency of rubidium to match.

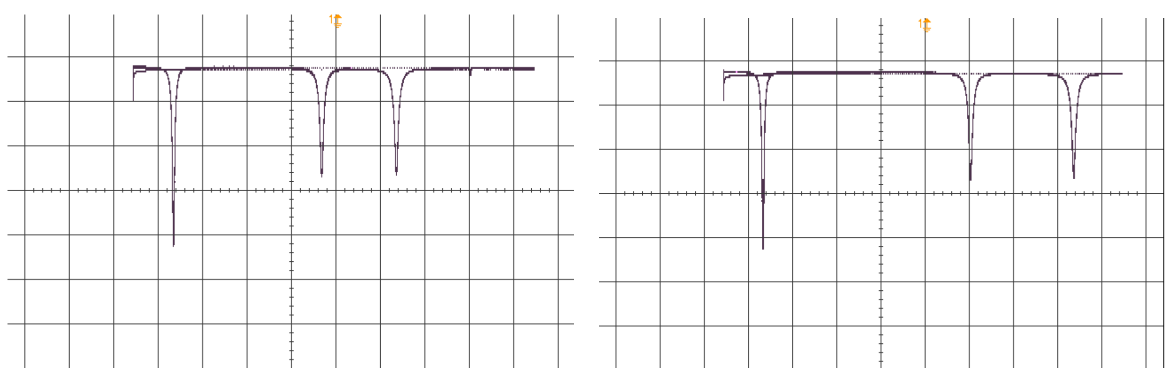

Figure 6: scope traces of resonant peaks observed at radio frequencies of 150 kHz (left) and 210 kHz (right). The traces clearly show the zero field resonances as well as the resonance dips for each of the two isotopes. The horizontal axis represents the horizontal sweep field, and the vertical axis represents the detected light intensity after the beam passed through the sample

Figure 6 contains the typical scope traces for observing rf resonance in rubidium. The peak on the farthest left represents the zero field resonance, and then the next peak represents the resonance in ^{87}Rb, with the final peak on the right of each trace representing the resonance in ^{85}Rb. The ^{87}Rb dip appears at a lower sweep field than ^{85}Rb because the energy splitting in the states is higher (as seen by the theoretical g-factors) meaning a lower magnetic field is needed for the same input rf frequency. The magnetic fields were measured as stated in section 2.3.

As can be seen in figure 6, changing the RF frequency changes the distance between the dips and the zero field dip, and the distance between the dips themselves. Increasing the RF frequency shifts the resonance dips to the right, representing a higher magnetic field strength. If the range of the sweep field is set to zero, the horizontal sweep field can be manually changed, and this causes the resonances to appear when the sweep field is set near the correct value.

RF Frequency [kHz] ± 0.02	**I (^{87}Rb) [mA]**	**I (^{85}Rb) [mA]**	**B (^{87}Rb) [G]**	**B (^{85}Rb) [G]**
51.62	296 ± 30	355 ± 30	0.0724 ± 0.02	0.108 ± 0.02
70.37	338 ± 30	421 ± 30	0.0978 ± 0.02	0.148 ± 0.02
91.02	387 ± 30	491 ± 30	0.127 ± 0.02	0.190 ± 0.02
110.1	430 ± 30	558 ± 30	0.153 ± 0.02	0.231 ± 0.02
130.0	477 ± 30	627 ± 30	0.182 ± 0.02	0.272 ± 0.02
150.38	525 ± 30	700 ± 30	0.211 ± 0.02	0.316 ± 0.02
171.04	572 ± 30	770 ± 30	0.239 ± 0.02	0.358 ± 0.02
189.98	616 ± 30	837 ± 30	0.266 ± 0.02	0.399 ± 0.02
205.05	650 ± 30	888 ± 30	0.286 ± 0.02	0.430 ± 0.02

Table 1: measured currents supplied to the horizontal sweep field at each resonance dip for various input radio frequencies

This data was plotted with a linear fit. The slope of the linear fit was used in equation 10 to calculate the g-factors for each isotope.

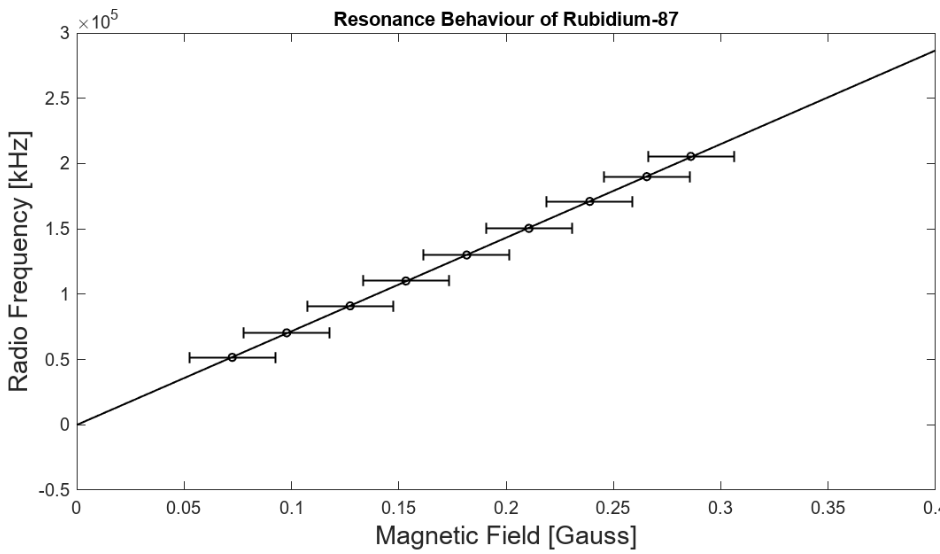

Figure 7: linear fit of resonance data from ^{87}Rb

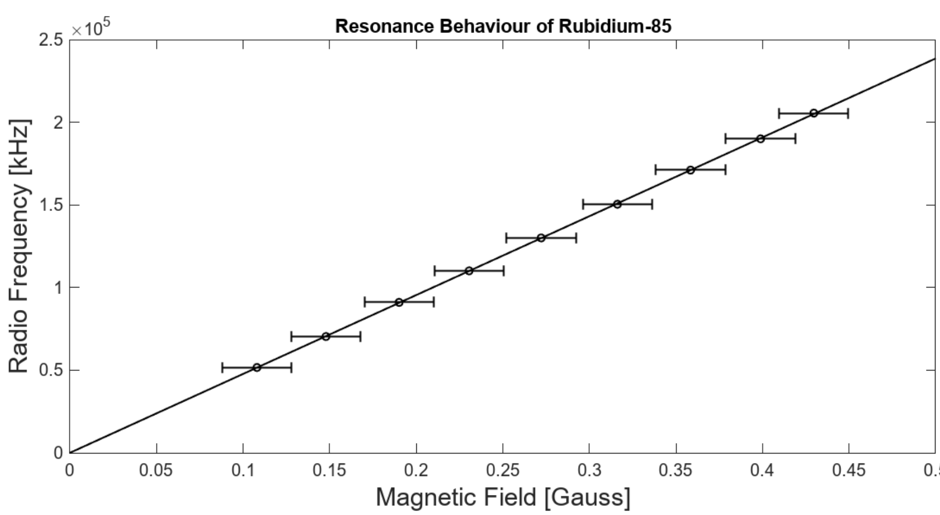

Figure 8: linear fit of resonance data from ^{85}Rb

For ^{87}Rb, the g-factor was calculated from the slope as $g_F = 0.512 \pm 0.01$, and for ^{85}Rb the g-factor was calculated to be $g_F = 0.341 \pm 0.01$. These values have percent differences of 2% with the theoretical values, and percent differences of 3% with other experimentally determined values [5].

3.3 Transient Phenomena

The properties of optical pumping in the time domain are of interetst, particularly the time it actually takes to optically pump our sample. This can be measured as described in section 2.3, by modulating the RF frequency. Turning on and off the RF field allows the sample to return to equilibrium and then be optically pumped again, which can be observed on the scope. The pumping curve is known to be exponential and therefore can be fit to a function of the form

$$f(t) = A(1 - e^{-t/\tau}) \tag{12}$$

The time constant τ is a measure of the optical pumping time.

The curves are shown in figure 9 as seen on the oscilloscope, without any fitting.

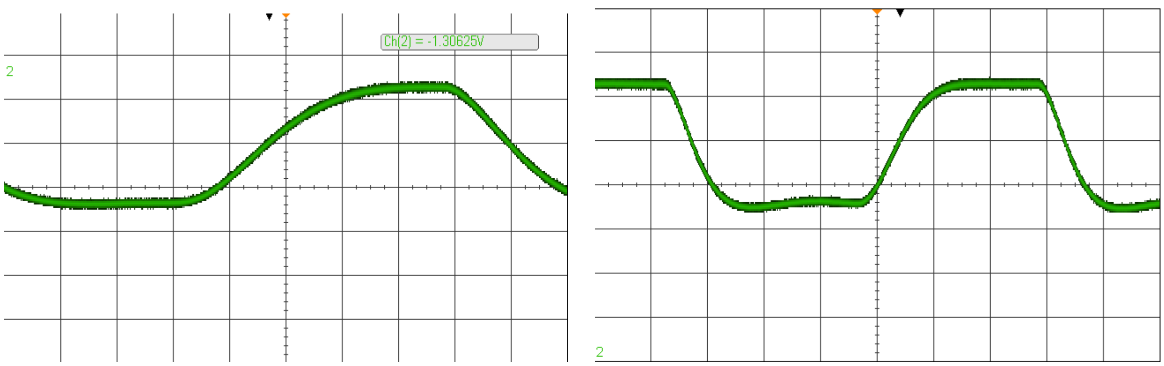

Figure 9: scope traces of the optical pumping of the rubidium isotopes ^{87}Rb (left) and ^{85}Rb demonstrating the exponential behaviour of these curves

These traces were fit with equation 12 and the resulting time constants were recorded.

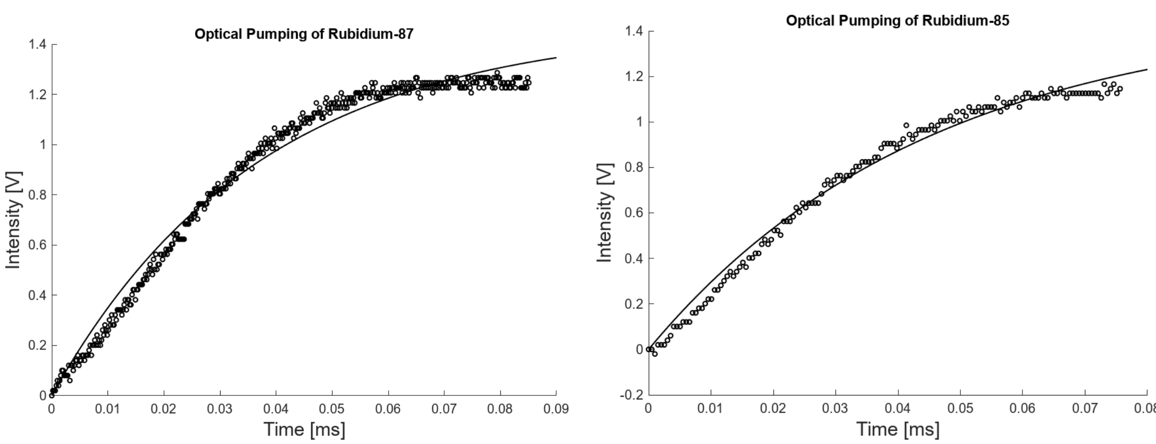

Figure 10: Caption

The time constants were $\tau_{87} = 36.9$ ms and $\tau_{85} = 44.8$ ms. These values make sense, as we expected the optical pumping time to be on the order of milliseconds.

4 Conclusion

The technique of optical pumping has applications in the exploration of fundamental particles and interactions, and our understanding of quantum physics and atomic dynamics. Probing the internal interactions of atoms has been an area of interest to physicists for years and the use of magnetic fields allows us to affect the dynamics of atoms and related particles. Optical pumping is a technique that utilizes these magnetic fields to force atoms into specific states, allowing us to observe and quantify certain state structures of the material. It is named 'Optical Pumping' as the material is 'pumped' with optical photons that force the atoms into a specific state, much like water is pumped from below ground to a higher elevation.

The states created due to the coupling of the nucleus and electron in an atom are called the hyperfine states, and their distribution is called the hyperfine structure. In this report we explored the use of optical pumping to quantify the energy splitting present in the hyperfine structure, and found the proportionality constants (called the g-factors) to within 3% of the theoretical and other experimental values.

In addition, we were able to measure the Earth's magnetic field to be 0.33 Gauss by finding the applied magnetic fields that caused zero field resonance in the sample.

The time it takes to optically pump this sample is on the order of milliseconds, and this was confirmed by measuring the time constant of the pumping curve as observed by modulating the RF field.

Optical pumping is a useful method to manipulate the energies of atoms and build a non-equilibrium system using magnetic fields. These systems have properties that scientists like to observe to further the understanding of quantum and nuclear physics.

References

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